

Parallel Trends as a Counterfactual Assumption

Teaching note

Schafhäutle and Veenman (2024), *The Accounting Review*

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The identifying assumption is untestable

“The identifying assumption underlying our difference-in-differences analyses is that the trends in the outcome variables would have moved in tandem in the counterfactual situation that Estimote had not added a firm to the platform. Although we cannot formally test this assumption, an alternative solution is to explore whether the treatment (Estimote) and control (non-Estimote) firms exhibit similar pretrends.”

Schafhäutle and Veenman (2024, p. 448)

- The assumption concerns a mean that is never observed; the evidence offered concerns means that are observed.

S&V estimate a two-way fixed effects DiD

$$Y_{it} = \alpha_i + \lambda_t + \delta Post_{it} + \varepsilon_{it} \quad (1)$$

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k D_i \mathbf{1}[t = k] + \varepsilon_{it} \quad (2)$$

Eq. (1): S&V Table 5. Firm and year-quarter effects; $Post_{it} = 1$ after Estimote coverage (p. 442).

Eq. (2): S&V Figure 5. Event years -3 to $+3$; year -1 omitted (p. 448).

$D_i = 1$ for firms that receive coverage at $t = 0$. $Y_{it}(1)$ and $Y_{it}(0)$ are the potential outcomes with and without coverage. Observed: $Y_{it} = Y_{it}(1)$ if $D_i = 1$ and $t \geq 0$; otherwise $Y_{it} = Y_{it}(0)$ (i.e., no anticipation).

■ The algebra that follows decomposes δ and β_k into a treatment effect and a term in $Y(0)$.

DiD imputes one mean from three others

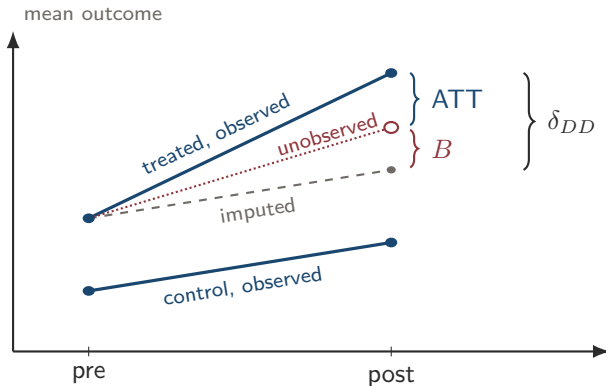
Observed: four group-by-period means.

Imputed: the treated pre-period mean plus the control group's change.

Never observed:

$E[Y_{post}(0) | D = 1]$, at a gap B from the imputed mean.

- The control group's change stands in for a treated-group change that is never observed.



The DiD estimand is the ATT plus a trend gap

In observed means, pooling the years before (pre) and after (post) treatment:

$$\delta_{DD} = \{E[Y_{post} | D=1] - E[Y_{pre} | D=1]\} - \{E[Y_{post} | D=0] - E[Y_{pre} | D=0]\} \quad (3)$$

Only treated firms after treatment reveal $Y(1)$. Add and subtract $E[Y_{post}(0) | D=1]$:

$$\delta_{DD} = \underbrace{E[Y_{post}(1) - Y_{post}(0) | D=1]}_{\text{ATT}} + \underbrace{E[Y_{post}(0) - Y_{pre}(0) | D=1] - E[Y_{post}(0) - Y_{pre}(0) | D=0]}_{B: \text{ gap in untreated trends}} \quad (4)$$

ATT: average treatment effect on the treated. With a balanced panel and one treatment date, $\hat{\delta}$ in eq. (1) is the sample analogue of δ_{DD} .

- Parallel trends is the statement $B = 0$, and B contains a mean that no dataset records.

Pre-trends identify only pre-period gaps

$$B_k \equiv E[Y_k(0) - Y_{-1}(0) \mid D=1] - E[Y_k(0) - Y_{-1}(0) \mid D=0]$$

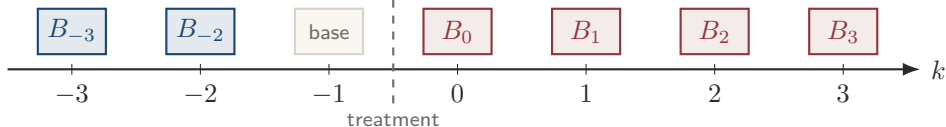
$$\beta_k = B_k \quad (k < 0)$$

$$\beta_k = \text{ATT}_k + B_k \quad (k \geq 0)$$

B_k : gap in untreated trends between year -1 and year k . ATT_k : the ATT in year k .

estimable: $Y = Y(0)$

assumed zero: treated $Y(0)$ unobserved



- Evidence that $B_k = 0$ before treatment neither implies nor is implied by $B_k = 0$ afterward (Kahn-Lang and Lang 2020).

Pre-trends and the assumption can disagree

```
* World 1: parallel trends holds
gen double y0_w1 = alpha + 0.1*k + eps
gen double y1_w1 = y0_w1 + path
* World 2: fails from k = 0
gen double y0_w2 = y0_w1 + D*path
gen double y1_w2 = y0_w2
* World 3: transitory pre-period shock
gen double y0_w3 = y0_w1 + D*blip
gen double y1_w3 = y0_w3 + path
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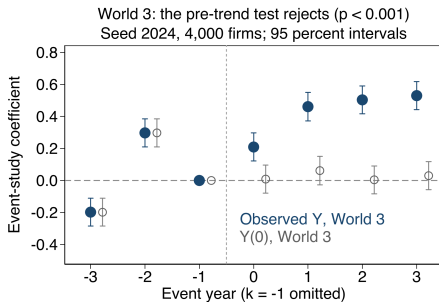
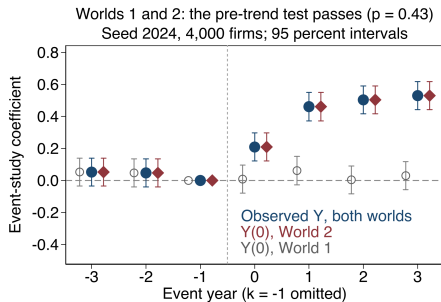
path = (0.2, 0.4, 0.5, 0.5) for $k \geq 0$; blip = (-0.25, +0.25) for $k = -3, -2$.

- The test passes where the assumption fails (World 2) and rejects where it holds (World 3).

	World 1	World 2	World 3
<i>Known only to the simulator</i>			
Parallel trends, $k \geq 0$	holds	fails	holds
Average ATT	0.40	0.00	0.40
<i>Computed from observed outcomes</i>			
Pre-trend F test, p	0.43	0.43	< 0.001
$\hat{\delta}$ on $Post(t)$	0.39 (16.2)	0.39 (16.2)	0.39 (16.2)
$\hat{\delta}$ minus ATT	-0.01	0.39	-0.01

Seed 2024; 4,000 firms. Pre-trend test: joint F test of $\beta_{-3} = \beta_{-2} = 0$. Worlds 1 and 2 share one observed panel.

Only $Y(0)$ regressions separate the worlds



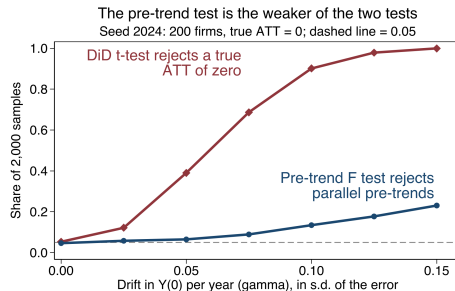
- Navy is what a researcher observes; gray and maroon require post-treatment $Y(0)$ of treated firms, known only in simulation.

Samples with a drift often pass the pre-test

ATT = 0. Relative to control firms, the $Y(0)$ of treated firms drifts by γ per year in every year, so that $E[\hat{\beta}_k] = \gamma(k + 1)$ and $E[\hat{\delta}] = 3.5\gamma$.

	All samples	Pre-test passed
Share of samples	1.00	0.87
Mean $\hat{\delta}$ (ATT = 0)	0.35	0.35
t -test rejects ATT = 0	0.90	0.90
Mean post $\hat{\beta}_k$ (bias 0.25)	0.26	0.28

Table: $\gamma = 0.10$; 2,000 samples of 200 firms.
`sv_parallel_trends_pretest.do`



- The pre-test confronts the two smallest deviations, -2γ and $-\gamma$, whereas $\hat{\delta}$ pools the full path (Roth 2022).

S&V Figure 5 is a falsification exercise

“In support of the parallel trends assumption, we find no evidence that suggests the pretrends are dissimilar for meet/beat incidence, announcement returns, and downward guidance frequency.”

Schafhäutle and Veenman (2024, p. 448)

Tested: $\hat{\beta}_{-3}$ and $\hat{\beta}_{-2}$, the estimated gaps before coverage begins.

Assumed: $B_k = 0$ once coverage begins, although coverage “is not random” (p. 450).

In defense of the design, additions are staggered over 20 quarters (p. 441), so that a confound would have to arrive in event time, firm by firm.

- Flat pre-trends are a test the design survived; they bear on the post-period counterfactual only under an assumption linking the periods.

Pre-trend tests fall short in several ways

Shortfall	Evidence in this note	Response in the literature
Passes although the assumption fails	World 2: observed data identical to World 1	Argue selection into treatment directly (Cunningham 2026)
Rejects although the assumption holds	World 3: $\hat{\delta}$ unbiased, $p < 0.001$	Argue why trends would be parallel (Kahn-Lang and Lang 2020)
Low power against drift	Rejections: 0.13 versus 0.90	Report pre-test power (Roth 2022)
Passing does not screen	0.90 among passing samples	Bound the violation by pre-period gaps (Rambachan and Roth 2023)
Functional form	Not simulated	Parallel in levels need not hold in logs (Roth and Sant'Anna 2023)

- The maroon shortfalls survive any sample size, because the test examines the wrong period; the navy ones concern power and specification.

Summary

- Parallel trends restricts the mean untreated outcome of treated firms after treatment, which no dataset records. The assumption is therefore not testable, and the DiD estimand equals the ATT plus the unobserved trend gap.
- Pre-trend graphs and tests are computed from observed outcomes before treatment. In the simulations they pass when the assumption fails, reject when it holds, and detect a drift considerably less often than the DiD t-test mistakes that drift for a treatment effect.
- Accordingly, flat pre-trends are perhaps best read as a falsification test that the design survived. The case for the assumption rests on institutional arguments about selection into treatment, supplemented by sensitivity analysis.